

Quantium Retail Strategy & Analytics – Task 2

Trial Store Analysis and Uplift Testing

The objective of this task is to assess whether the new store layout trial had a measurable impact on chip category performance.

The trial was conducted in stores 77, 86 and 88.

To assess performance, each trial store will be compared against a similar control store based on pre-trial store behaviour.

The analysis focuses on:

- monthly total sales revenue
- monthly number of customers
- monthly transactions per customer
- selecting suitable control stores
- comparing trial and control performance during the trial period
- identifying whether the trial should be rolled out more broadly

1. Import Libraries

The required Python libraries are imported for data manipulation, statistical analysis and visualisation.

```
In [1]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

2. Load Dataset

The cleaned output from Task 1 is used as the starting point for Task 2.

This avoids repeating earlier cleaning steps and allows the analysis to focus on store-level trial performance.

```
In [2]: try:
data=pd.read_csv('/content/merged_data.csv')
print(f'Dataset shape{data.shape}')
except Exception as e:
print(f'Error loading the dataset: {e}')
```

Dataset shape(264834, 13)

3. Initial Data Check

Before creating store-level metrics, the dataset structure is reviewed to confirm column types, missing values and key fields required for the trial analysis.

In [3]: `data.info()`

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 264834 entries, 0 to 264833
Data columns (total 13 columns):
#   Column                Non-Null Count  Dtype
---  -
0   DATE                  264834 non-null object
1   STORE_NBR             264834 non-null int64
2   LYLTY_CARD_NBR        264834 non-null int64
3   TXN_ID                264834 non-null int64
4   PROD_NBR              264834 non-null int64
5   PROD_NAME             264834 non-null object
6   PROD_QTY              264834 non-null int64
7   TOT_SALES             264834 non-null float64
8   PACK_SIZE             264834 non-null int64
9   BRAND                 264834 non-null object
10  LIFESTAGE              264834 non-null object
11  PREMIUM_CUSTOMER      264834 non-null object
12  MULTIPACK             264834 non-null bool
dtypes: bool(1), float64(1), int64(6), object(5)
memory usage: 24.5+ MB
```

4. Convert Date Column

The date column is converted into datetime format so that monthly store performance can be analysed correctly.

In [4]: `data['DATE']=pd.to_datetime(data['DATE'])`

In [5]: `print(data['DATE'].dtype)`

```
datetime64[ns]
```

Initial Observations

The dataset contains transaction-level chip purchase data across multiple stores.

Key variables relevant for the trial analysis include:

- STORE_NBR for store-level comparison
- DATE for monthly trend analysis
- TOT_SALES for revenue measurement
- TXN_ID for transaction counts
- LYLTY_CARD_NBR for customer counts

The analysis will focus on store-level performance trends over time.

5. Store Coverage Overview

The trial analysis is store-based, so the first step is to understand how many stores are present in the dataset and how transaction volume varies by store.

```
In [6]: data['STORE_NBR'].nunique()
```

```
Out[6]: 272
```

```
In [7]: data['STORE_NBR'].value_counts().sort_values()
```

```
Out[7]:
```

STORE_NBR	count
92	1
76	1
211	2
252	2
11	2
...	...
237	1785
165	1819
93	1832
88	1873
226	2020

272 rows × 1 columns

dtype: int64

Store Distribution Review

Transaction activity varies considerably across stores.

Some stores contain very limited observations while others contribute substantially more transactions.

This reinforces the importance of selecting comparable control stores during the trial analysis.

6. Create Monthly Period Column

Task 2 requires store performance to be compared month by month.

The transaction-level dates are therefore converted into a monthly period column.

```
In [8]: #to extract month and year out of dates
data['MONTH_ID'] = data['DATE'].dt.to_period('M')
```

```
In [9]: data['MONTH_ID'].head(5)
```

```
Out[9]:
```

	MONTH_ID
0	2018-10
1	2019-05
2	2019-05
3	2018-08
4	2018-08

dtype: period[M]

7. Create Monthly Store-Level Metrics

The transaction-level dataset is aggregated into monthly store-level metrics.

For each store and month, the analysis calculates:

- total sales revenue
- number of unique customers
- number of transactions
- transactions per customer

This monthly store-level table will be used to identify suitable control stores and assess trial performance.

```
In [10]: monthly_store_metrics = data.groupby(  
        ['STORE_NBR', 'MONTH_ID']  
        ).agg(  
            total_sales=('TOT_SALES', 'sum'),  
            unique_customers=('LYLTY_CARD_NBR', 'nunique'),  
            total_transactions=('TXN_ID', 'nunique')  
        ).reset_index()
```

```
In [11]: monthly_store_metrics['transactions_per_customer'] = (  
        monthly_store_metrics['total_transactions'] /  
        monthly_store_metrics['unique_customers']  
        )
```

```
In [12]: monthly_store_metrics.head()
```

Out[12]:

	STORE_NBR	MONTH_ID	total_sales	unique_customers	total_transactions	transactio
0	1	2018-07	206.9	49	52	
1	1	2018-08	176.1	42	43	
2	1	2018-09	278.8	59	62	
3	1	2018-10	188.1	44	45	
4	1	2018-11	192.6	46	47	



Monthly Store-Level Metrics

The transaction-level dataset has now been aggregated into monthly store-level metrics.

Each row now represents:

- one store
- during one month
- with summarized performance indicators

This structure is required for comparing historical store behaviour and identifying suitable control stores.

8. Check Monthly Coverage by Store

Only stores with sales data across all months should be considered as possible control stores.

Stores with incomplete monthly records may produce unreliable comparisons.

```
In [13]: store_month_counts = monthly_store_metrics.groupby(
          'STORE_NBR'
        )['MONTH_ID'].nunique().sort_values()
```

```
In [14]: store_month_counts.head()
```

Out[14]:

	MONTH_ID
STORE_NBR	
85	1
92	1
76	1
31	2
206	2

dtype: int64

```
In [15]: store_month_counts.tail()
```

```
Out[15]:
```

	MONTH_ID
STORE_NBR	
21	12
20	12
19	12
18	12
17	12

dtype: int64

```
In [16]: complete_stores = store_month_counts[store_month_counts == 12].index
```

```
In [17]: monthly_store_metrics_complete = monthly_store_metrics[
    monthly_store_metrics['STORE_NBR'].isin(complete_stores)
]
```

```
In [18]: monthly_store_metrics_complete['STORE_NBR'].nunique()
```

```
Out[18]: 260
```

Filtering Stores with Complete Monthly Data

Only stores with observations across all 12 months were retained for analysis.

Stores with incomplete monthly data may distort similarity comparisons and produce unreliable trial assessments.

9. Define Trial and Pre-Trial Periods

The trial period ran from February 2019 to April 2019.

The pre-trial period is used to identify control stores that had similar historical performance before the layout change.

```
In [19]: trial_stores = [77, 86, 88]

pre_trial_data = monthly_store_metrics_complete[
    monthly_store_metrics_complete['MONTH_ID'] < '2019-02'
]

trial_period_data = monthly_store_metrics_complete[
    (monthly_store_metrics_complete['MONTH_ID'] >= '2019-02') &
    (monthly_store_metrics_complete['MONTH_ID'] <= '2019-04')
]
```

```
In [20]: pre_trial_data.head()
```

Out[20]:

	STORE_NBR	MONTH_ID	total_sales	unique_customers	total_transactions	transactio
0	1	2018-07	206.9	49	52	
1	1	2018-08	176.1	42	43	
2	1	2018-09	278.8	59	62	
3	1	2018-10	188.1	44	45	
4	1	2018-11	192.6	46	47	

10. Control Store Selection Method

Control stores are selected using pre-trial performance only.

A suitable control store should have similar historical sales and customer behaviour to the trial store before the layout change.

Similarity will be assessed using:

- correlation in monthly sales trends
- correlation in monthly customer trends
- similarity in average monthly sales levels
- similarity in average monthly customer counts

```
In [21]: def calculate_control_scores(trial_store, pre_trial_data, trial_stores):

    candidate_stores = pre_trial_data[
        ~pre_trial_data['STORE_NBR'].isin(trial_stores)
    ][['STORE_NBR']].unique()

    scores = []

    trial_data = pre_trial_data[
        pre_trial_data['STORE_NBR'] == trial_store
    ].sort_values('MONTH_ID').reset_index(drop=True)

    for control_store in candidate_stores:

        control_data = pre_trial_data[
            pre_trial_data['STORE_NBR'] == control_store
        ].sort_values('MONTH_ID').reset_index(drop=True)

        sales_corr = trial_data['total_sales'].corr(control_data['total_sales'])
        customer_corr = trial_data['unique_customers'].corr(control_data['unique

        sales_diff = abs(
            trial_data['total_sales'].mean() - control_data['total_sales'].mean(
        )

        customer_diff = abs(
            trial_data['unique_customers'].mean() - control_data['unique_custome
```

```

        scores.append({
            'trial_store': trial_store,
            'control_store': control_store,
            'sales_corr': sales_corr,
            'customer_corr': customer_corr,
            'sales_diff': sales_diff,
            'customer_diff': customer_diff
        })

    scores_df = pd.DataFrame(scores)

    scores_df['sales_mag_score'] = 1 - (
        (scores_df['sales_diff'] - scores_df['sales_diff'].min()) /
        (scores_df['sales_diff'].max() - scores_df['sales_diff'].min())
    )

    scores_df['customer_mag_score'] = 1 - (
        (scores_df['customer_diff'] - scores_df['customer_diff'].min()) /
        (scores_df['customer_diff'].max() - scores_df['customer_diff'].min())
    )

    scores_df['final_score'] = (
        scores_df['sales_corr'] * 0.25 +
        scores_df['customer_corr'] * 0.25 +
        scores_df['sales_mag_score'] * 0.25 +
        scores_df['customer_mag_score'] * 0.25
    )

    return scores_df.sort_values('final_score', ascending=False)

```

```

In [22]: #select the control
scores_77 = calculate_control_scores(77, pre_trial_data, trial_stores)
scores_86 = calculate_control_scores(86, pre_trial_data, trial_stores)
scores_88 = calculate_control_scores(88, pre_trial_data, trial_stores)

```

11. Calculate Final Similarity Score

Control stores are ranked using a combined similarity measure.

The final score considers:

- similarity in monthly sales movement
- similarity in customer trends
- closeness in sales magnitude
- closeness in customer magnitude

Higher scores indicate stronger similarity to the trial store.

```

In [23]: scores_77['final_score'] = (
    scores_77['sales_corr'] +
    scores_77['customer_corr'] +
    scores_77['sales_mag_score'] +
    scores_77['customer_mag_score']
) / 4

scores_77 = scores_77.sort_values(

```

```

    'final_score',
    ascending=False
)

scores_77.head(10)

```

Out[23]:

	trial_store	control_store	sales_corr	customer_corr	sales_diff	customer_diff	sales
218	77	233	0.903774	0.990358	5.600000	0.142857	
38	77	41	0.783232	0.844219	45.342857	0.142857	
15	77	17	0.842668	0.747308	155.614286	1.000000	
238	77	254	0.577108	0.916208	101.057143	6.142857	
156	77	167	0.657110	0.717913	53.485714	2.000000	
105	77	115	0.689159	0.718882	87.657143	2.857143	
78	77	84	0.684348	0.858571	219.928571	7.428571	
46	77	50	0.763866	0.607391	28.085714	6.285714	
101	77	111	0.519473	0.685926	38.428571	0.142857	
249	77	265	0.639759	0.573460	37.557143	1.000000	



In [24]:

```

scores_86['final_score'] = (
    scores_86['sales_corr'] +
    scores_86['customer_corr'] +
    scores_86['sales_mag_score'] +
    scores_86['customer_mag_score']
) / 4

scores_86 = scores_86.sort_values(
    'final_score',
    ascending=False
)

scores_86.head(10)

```

Out[24]:

	trial_store	control_store	sales_corr	customer_corr	sales_diff	customer_diff	sale
144	86	155	0.877882	0.942876	26.978571	0.000000	
99	86	109	0.788300	0.770778	16.921429	0.428571	
127	86	138	0.759864	0.749701	29.021429	3.000000	
104	86	114	0.734415	0.855339	68.757143	6.285714	
210	86	225	0.617533	0.733791	4.178571	0.428571	
207	86	222	0.795075	0.458238	20.835714	5.571429	
161	86	172	0.524475	0.571024	0.921429	3.285714	
136	86	147	0.545463	0.641090	60.657143	8.000000	
52	86	56	0.551441	0.703549	154.850000	15.428571	
71	86	75	0.706307	0.578150	263.378571	7.714286	



In [25]:

```

scores_88['final_score'] = (
    scores_88['sales_corr'] +
    scores_88['customer_corr'] +
    scores_88['sales_mag_score'] +
    scores_88['customer_mag_score']
) / 4

scores_88 = scores_88.sort_values(
    'final_score',
    ascending=False
)

scores_88.head(10)

```

Out[25]:

	trial_store	control_store	sales_corr	customer_corr	sales_diff	customer_diff	sale
222	88	237	0.308479	0.947326	2.085714	0.714286	
167	88	178	0.731857	0.939466	405.585714	22.142857	
65	88	69	0.450029	0.815792	389.478571	16.142857	
103	88	113	0.495763	0.862632	422.871429	26.714286	
189	88	201	0.492735	0.511302	169.742857	9.857143	
112	88	123	0.399761	0.627925	182.985714	12.142857	
191	88	203	0.508001	0.285063	62.842857	5.428571	
241	88	257	0.311789	0.703602	416.885714	19.571429	
6	88	7	0.649657	0.193914	295.728571	20.571429	
24	88	26	0.300009	0.321523	131.457143	8.857143	



12. Initial Control Store Selection Based on Similarity Scores

Top-ranked control store candidates were reviewed using both similarity scores and visual trend comparison.

The objective is to select stores with:

- similar monthly sales movement
- similar customer behaviour
- comparable sales magnitude

Visual validation is important because some stores may have similar averages but very different sales trends over time.

```
In [26]: control_store_77 = 233  
control_store_86 = 155  
control_store_88 = 237
```

13. Visual Comparison of Trial and Control Stores

The following visualisations compare monthly sales performance between each trial store and its selected control store.

This helps validate whether the control store demonstrates similar behaviour prior to the trial period.

```
In [27]: def plot_store_comparison(trial_store, control_store, metric='total_sales'):  
  
    trial_data = monthly_store_metrics_complete[  
        monthly_store_metrics_complete['STORE_NBR'] == trial_store  
    ]  
  
    control_data = monthly_store_metrics_complete[  
        monthly_store_metrics_complete['STORE_NBR'] == control_store  
    ]  
  
    plt.figure(figsize=(12,6))  
  
    plt.plot(  
        trial_data['MONTH_ID'].astype(str),  
        trial_data[metric],  
        marker='o',  
        label=f'Trial Store {trial_store}'  
    )  
  
    plt.plot(  
        control_data['MONTH_ID'].astype(str),  
        control_data[metric],  
        marker='o',  
        label=f'Control Store {control_store}'  
    )
```

```

plt.axvspan('2019-02', '2019-04', color='lightgrey', alpha=0.4)

plt.title(f'{metric} Comparison')
plt.xlabel('Month')
plt.ylabel(metric)

plt.xticks(rotation=45)
plt.legend()
plt.savefig(
    f'{trial_store}_vs_{control_store}.png',
    dpi=300,
    bbox_inches='tight'
)

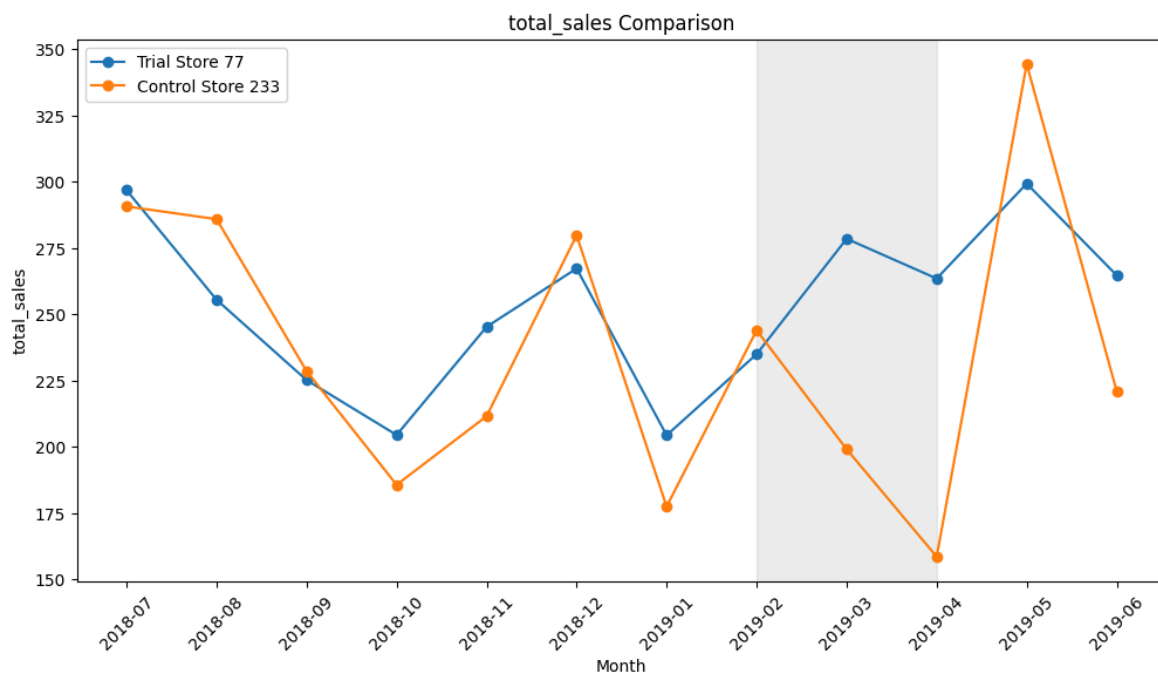
plt.show()

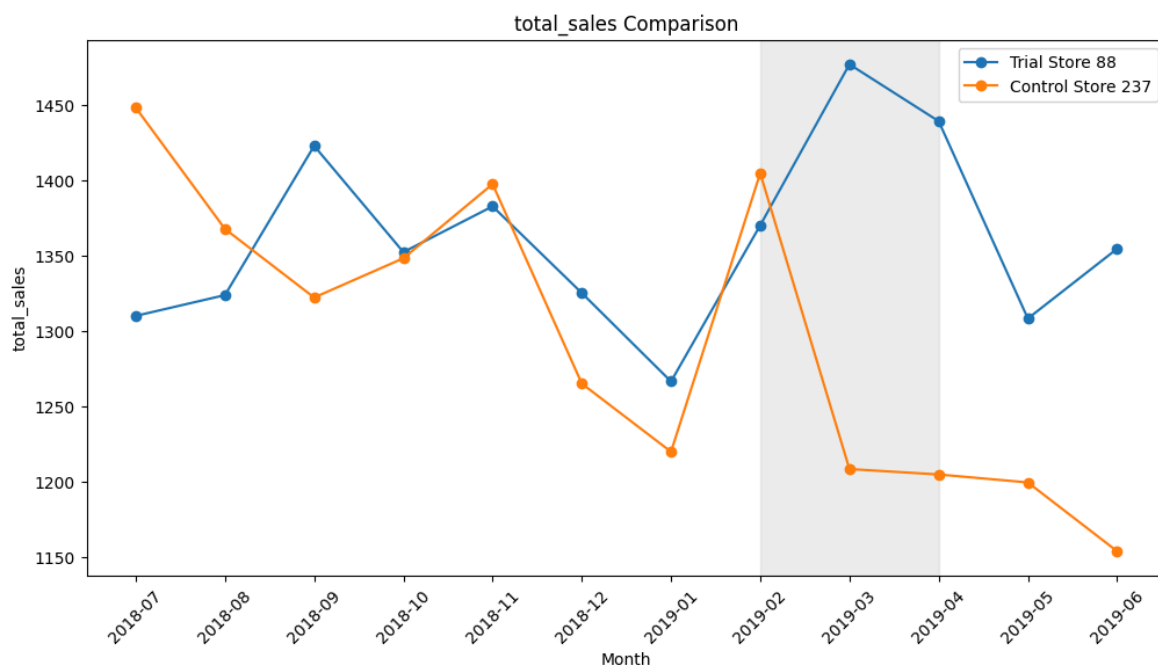
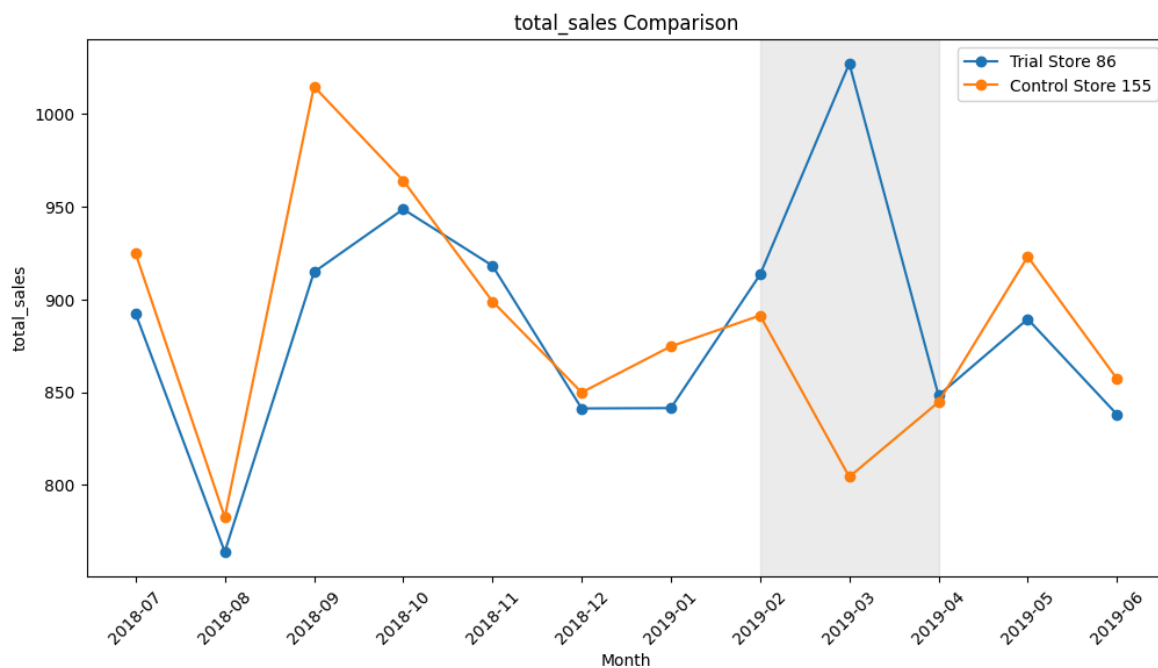
```

```

In [28]: plot_store_comparison(77, 233)
plot_store_comparison(86, 155)
plot_store_comparison(88, 237)

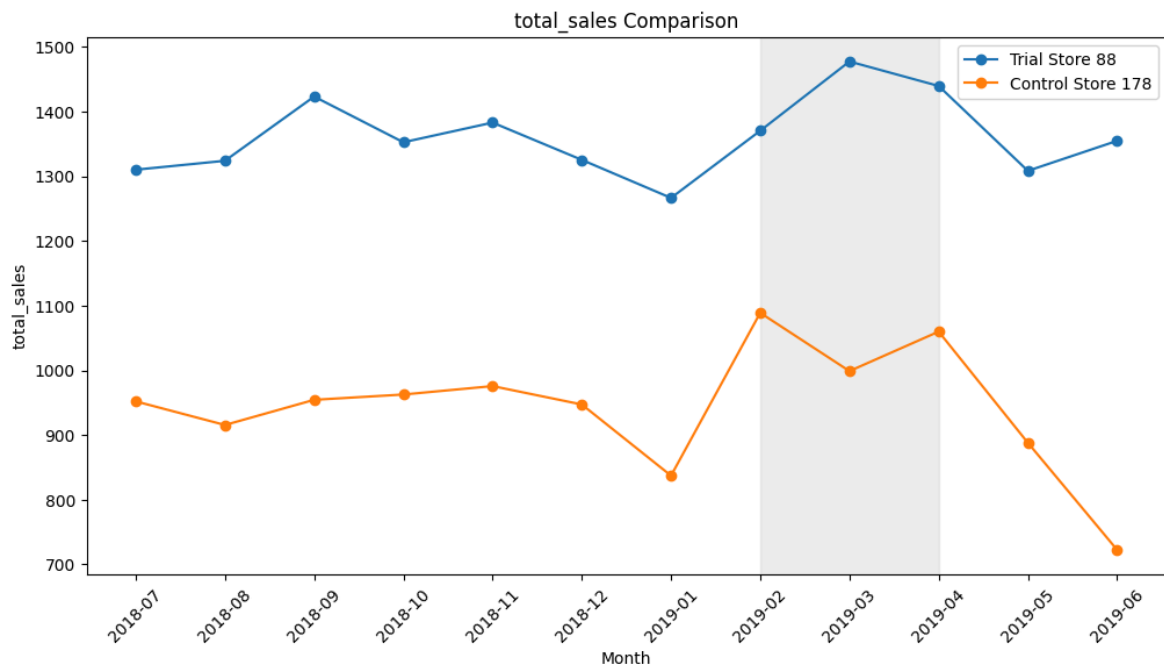
```





Additional Store 88 Control Validation

```
In [29]: plot_store_comparison(88, 178)
```



Final Control Store Decision

Store 237 was initially selected for Store 88 based on the combined similarity score. However, visual validation showed weaker alignment in monthly sales movement during the pre-trial period.

Additional candidates were reviewed, and Store 178 was selected as the final control store for Store 88 because it showed a more reasonable historical sales pattern for comparison.

The final control stores used for uplift testing are:

- Store 77 → Store 233
- Store 86 → Store 155
- Store 88 → Store 178

14. Trial Period Uplift Assessment

The selected control stores were scaled to align with the pre-trial performance of the trial stores.

This section evaluates whether trial stores experienced significantly different sales performance during the trial period from February 2019 to April 2019.

```
In [30]: final_control_store_77 = 233
         final_control_store_86 = 155
         final_control_store_88 = 178
```

```
In [31]: #trial pairs
         store_pairs = {
             77: 233,
             86: 155,
```

```
88: 178
}
```

Scaling Control Stores

Control stores may operate at different absolute sales levels compared to trial stores.

To enable fair comparison, control store sales are scaled using the ratio of pre-trial sales between the trial and control stores.

This adjustment ensures that differences observed during the trial period are more likely attributable to the trial layout rather than baseline store size differences.

```
In [32]: def calculate_scaling_factor(trial_store, final_control_store):
```

```
    trial_sales = pre_trial_data[
        pre_trial_data['STORE_NBR'] == trial_store
    ]['total_sales'].sum()

    control_sales = pre_trial_data[
        pre_trial_data['STORE_NBR'] == final_control_store
    ]['total_sales'].sum()

    scaling_factor = trial_sales / control_sales

    return scaling_factor
```

```
In [33]: # calculating scaling factor
```

```
scaling_77 = calculate_scaling_factor(77, 233)
scaling_86 = calculate_scaling_factor(86, 155)
scaling_88 = calculate_scaling_factor(88, 178)

print(scaling_77)
print(scaling_86)
print(scaling_88)
```

```
1.023617303289553
0.9700651481287743
1.4338146535258616
```

```
In [34]: def scale_control_store(trial_store, final_control_store, scaling_factor):
```

```
    control_data = monthly_store_metrics_complete[
        monthly_store_metrics_complete['STORE_NBR'] == final_control_store
    ].copy()

    control_data['scaled_sales'] = (
        control_data['total_sales'] * scaling_factor
    )

    control_data['trial_store'] = trial_store

    return control_data
```

```
In [35]: scaled_control_77 = scale_control_store(77, 233, scaling_77)
scaled_control_86 = scale_control_store(86, 155, scaling_86)
scaled_control_88 = scale_control_store(88, 178, scaling_88)
```

Comparing Trial Stores Against Scaled Controls

The scaled control stores are compared against trial stores across the full observation period.

The shaded region represents the trial period from February 2019 to April 2019.

If the trial layout had a positive impact, trial store performance should diverge positively from the scaled control store during the trial period.

```
In [36]: def plot_scaled_comparison(trial_store, final_control_store, scaled_control_data

        trial_data = monthly_store_metrics_complete[
            monthly_store_metrics_complete['STORE_NBR'] == trial_store
        ]

        plt.figure(figsize=(12,6))

        # Trial store sales
        plt.plot(
            trial_data['MONTH_ID'].astype(str),
            trial_data['total_sales'],
            marker='o',
            label=f'Trial Store {trial_store}'
        )

        # Scaled control sales
        plt.plot(
            scaled_control_data['MONTH_ID'].astype(str),
            scaled_control_data['scaled_sales'],
            marker='o',
            label=f'Scaled Control Store {final_control_store}'
        )

        # Highlight trial period
        plt.axvspan('2019-02', '2019-04', alpha=0.2)

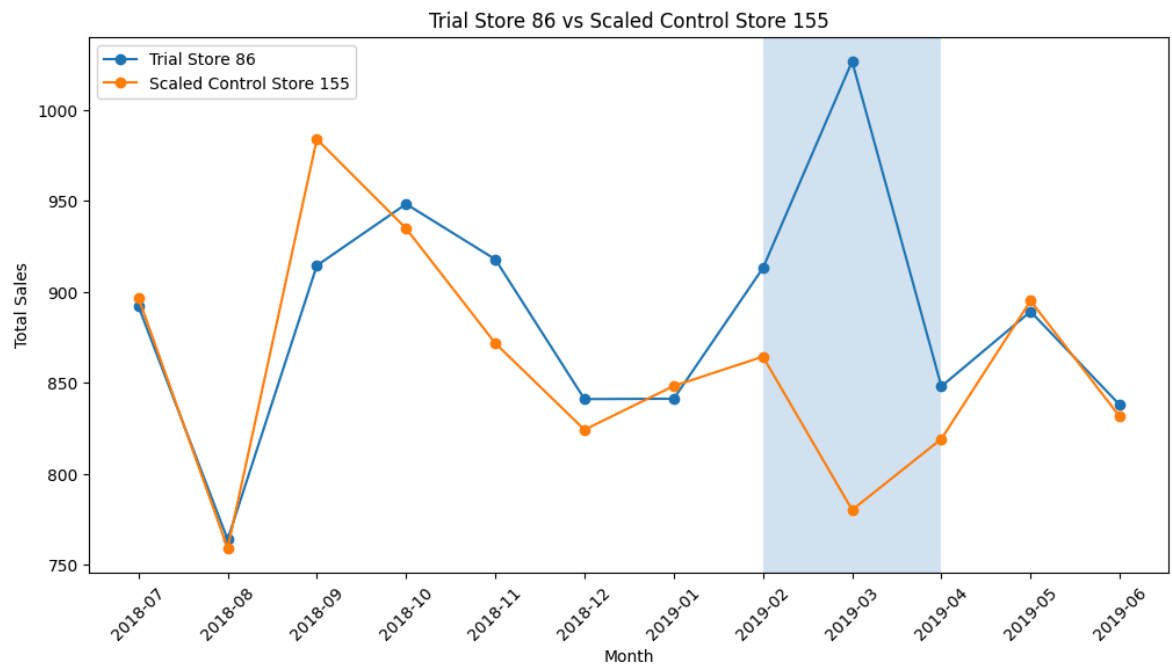
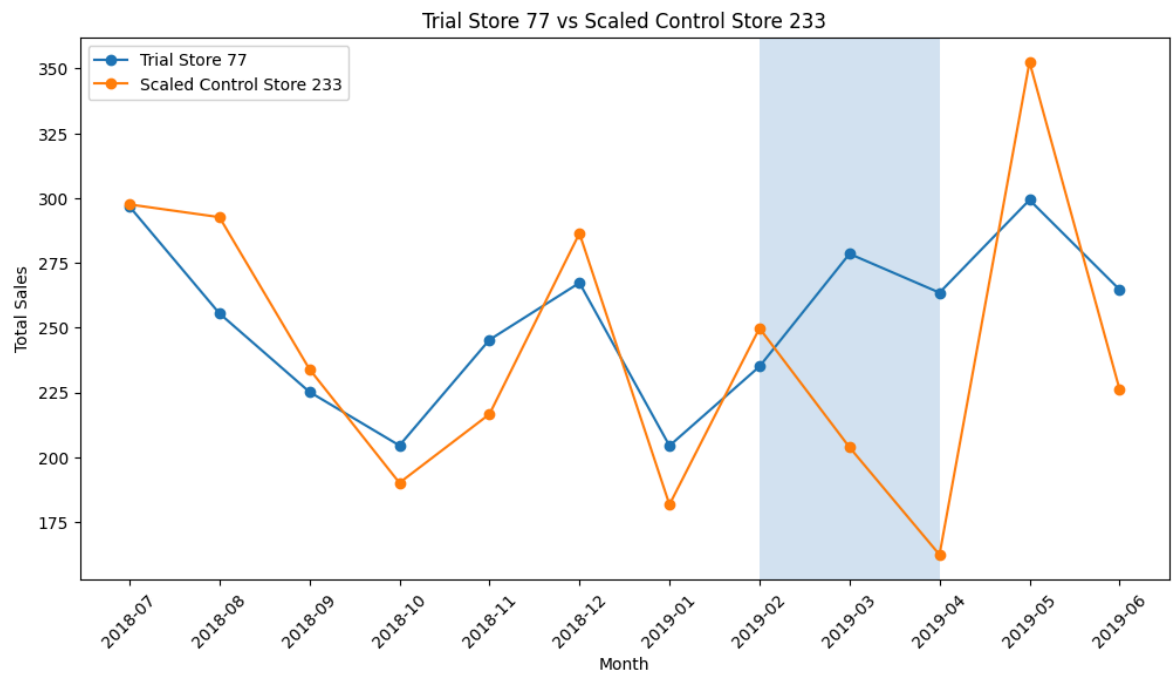
        plt.title(f'Trial Store {trial_store} vs Scaled Control Store {final_control}')
        plt.xlabel('Month')
        plt.ylabel('Total Sales')

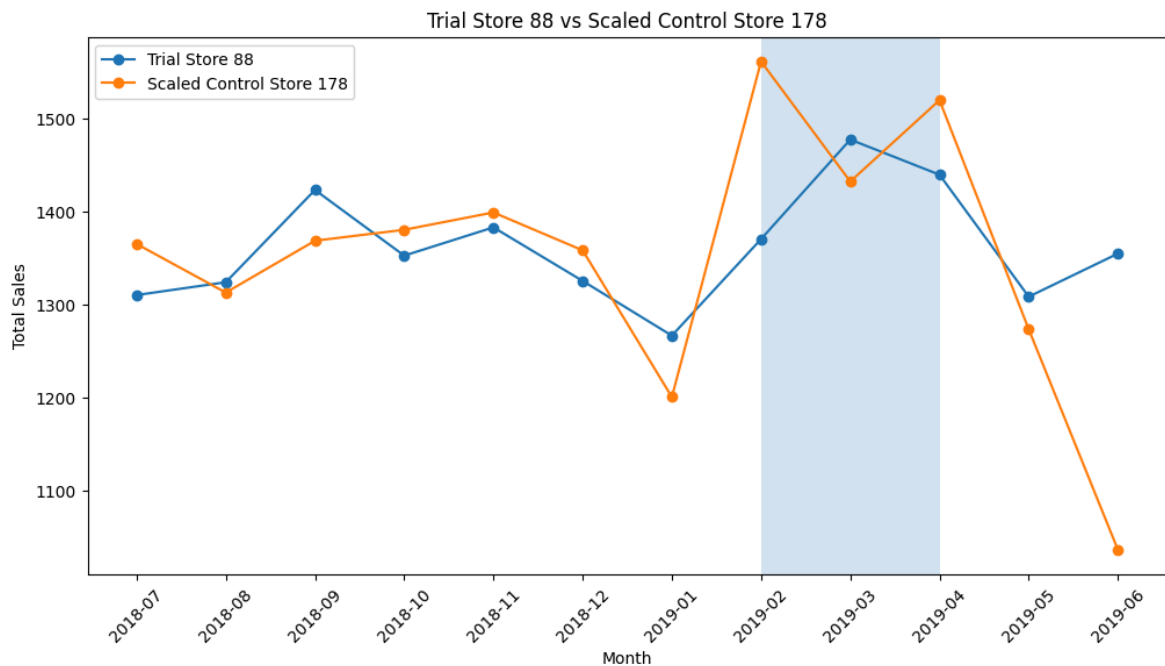
        plt.xticks(rotation=45)

        plt.legend()

        plt.show()
```

```
In [37]: plot_scaled_comparison(77, 233, scaled_control_77)
        plot_scaled_comparison(86, 155, scaled_control_86)
        plot_scaled_comparison(88, 178, scaled_control_88)
```





Initial Trial Period Observations

After scaling, the control stores provide a more appropriate benchmark for assessing trial performance.

Trial Store 77 shows noticeable sales uplift during the trial period relative to its scaled control store.

Trial Store 86 demonstrates weaker evidence of uplift, with sales movements remaining relatively similar to the control store.

Trial Store 88 does not show clear positive divergence during the trial period, and control matching quality was weaker compared to the other trial stores.

Statistical testing will now be conducted to determine whether observed differences are statistically significant.

15. Trial Period Sales Uplift

This section calculates sales uplift during the trial period by comparing actual trial store sales against expected sales from scaled control stores.

```
In [38]: def calculate_percentage_uplift(trial_store, scaled_control_data):

    # Trial store data
    trial_data = monthly_store_metrics_complete[
        monthly_store_metrics_complete['STORE_NBR'] == trial_store
    ][['MONTH_ID', 'total_sales']]

    # Merge with scaled control
    comparison = trial_data.merge(
        scaled_control_data[['MONTH_ID', 'scaled_sales']],
        on='MONTH_ID'
    )
```

```

# Calculate uplift %
comparison['uplift_percentage'] = (
    (comparison['total_sales'] - comparison['scaled_sales'])
    / comparison['scaled_sales']
) * 100

return comparison

```

```

In [39]: uplift_77 = calculate_percentage_uplift(77, scaled_control_77)
         uplift_86 = calculate_percentage_uplift(86, scaled_control_86)
         uplift_88 = calculate_percentage_uplift(88, scaled_control_88)

```

```

In [40]: uplift_77.head()

```

```

Out[40]:
```

	MONTH_ID	total_sales	scaled_sales	uplift_percentage
0	2018-07	296.8	297.565550	-0.257271
1	2018-08	255.5	292.652187	-12.694997
2	2018-09	225.2	233.998916	-3.760238
3	2018-10	204.5	190.085733	7.583035
4	2018-11	245.3	216.597421	13.251579

```

In [41]: trial_uplift_77 = uplift_77[
         (uplift_77['MONTH_ID'] >= '2019-02') &
         (uplift_77['MONTH_ID'] <= '2019-04')
         ]

         trial_uplift_86 = uplift_86[
         (uplift_86['MONTH_ID'] >= '2019-02') &
         (uplift_86['MONTH_ID'] <= '2019-04')
         ]

         trial_uplift_88 = uplift_88[
         (uplift_88['MONTH_ID'] >= '2019-02') &
         (uplift_88['MONTH_ID'] <= '2019-04')
         ]

```

```

In [42]: print(trial_uplift_77['uplift_percentage'].mean())
         print(trial_uplift_86['uplift_percentage'].mean())
         print(trial_uplift_88['uplift_percentage'].mean())

```

```

31.01646728364261
13.580947268464397
-4.785340331097365

```

Trial Period Uplift Summary

Average sales uplift during the trial period was:

- Store 77: approximately +31.0%
- Store 86: approximately +13.6%
- Store 88: approximately -4.8%

Store 77 shows the strongest positive uplift compared with its scaled control store. Store 86 shows a smaller positive uplift, while Store 88 does not show positive uplift based on the selected control store.

16. Statistical Comparison Against Historical Performance

The standard deviation of pre-trial uplift is used as a benchmark for normal variation between each trial store and its scaled control store.

Trial-period uplift is then compared against this historical variation. A larger t-value indicates that observed uplift is less likely to be explained by normal pre-trial fluctuation.

```
In [43]: def calculate_significance_table(uplift_data):
    pre_trial = uplift_data[uplift_data['MONTH_ID'] < '2019-02'].copy()
    trial_period = uplift_data[
        (uplift_data['MONTH_ID'] >= '2019-02') &
        (uplift_data['MONTH_ID'] <= '2019-04')
    ].copy()

    std_dev = pre_trial['uplift_percentage'].std()

    trial_period['t_value'] = trial_period['uplift_percentage'] / std_dev

    return trial_period, std_dev
```

```
In [44]: sig_77, std_77 = calculate_significance_table(uplift_77)
sig_86, std_86 = calculate_significance_table(uplift_86)
sig_88, std_88 = calculate_significance_table(uplift_88)
```

```
In [45]: sig_77
```

```
Out[45]:
```

	MONTH_ID	total_sales	scaled_sales	uplift_percentage	t_value
7	2019-02	235.0	249.762622	-5.910661	-0.593520
8	2019-03	278.5	203.802205	36.652103	3.680430
9	2019-04	263.5	162.345704	62.307960	6.256669

```
In [46]: sig_86
```

```
Out[46]:
```

	MONTH_ID	total_sales	scaled_sales	uplift_percentage	t_value
7	2019-02	913.2	864.522060	5.630619	1.494114
8	2019-03	1026.8	780.320405	31.586973	8.381769
9	2019-04	848.2	819.317024	3.525250	0.935444

```
In [47]: sig_88
```

Out[47]:

	MONTH_ID	total_sales	scaled_sales	uplift_percentage	t_value
7	2019-02	1370.2	1561.137395	-12.230659	-3.476645
8	2019-03	1477.2	1432.094076	3.149648	0.895308
9	2019-04	1439.4	1519.556770	-5.275010	-1.499456

In [48]:

```
print("Store 77 pre-trial standard deviation:", std_77)
print("Store 86 pre-trial standard deviation:", std_86)
print("Store 88 pre-trial standard deviation:", std_88)
```

Store 77 pre-trial standard deviation: 9.958646884078389
 Store 86 pre-trial standard deviation: 3.768532790008376
 Store 88 pre-trial standard deviation: 3.5179492474944594

17. Trial Performance Visualization

The following visualisations compare monthly percentage uplift during the trial period relative to historical variation thresholds.

The dotted horizontal lines represent approximate significance boundaries.

In [49]:

```
def plot_trial_significance(sig_data, trial_store):

    plt.figure(figsize=(10,5))

    plt.plot(
        sig_data['MONTH_ID'].astype(str),
        sig_data['t_value'],
        marker='o'
    )

    plt.axhline(2, linestyle='--')
    plt.axhline(-2, linestyle='--')

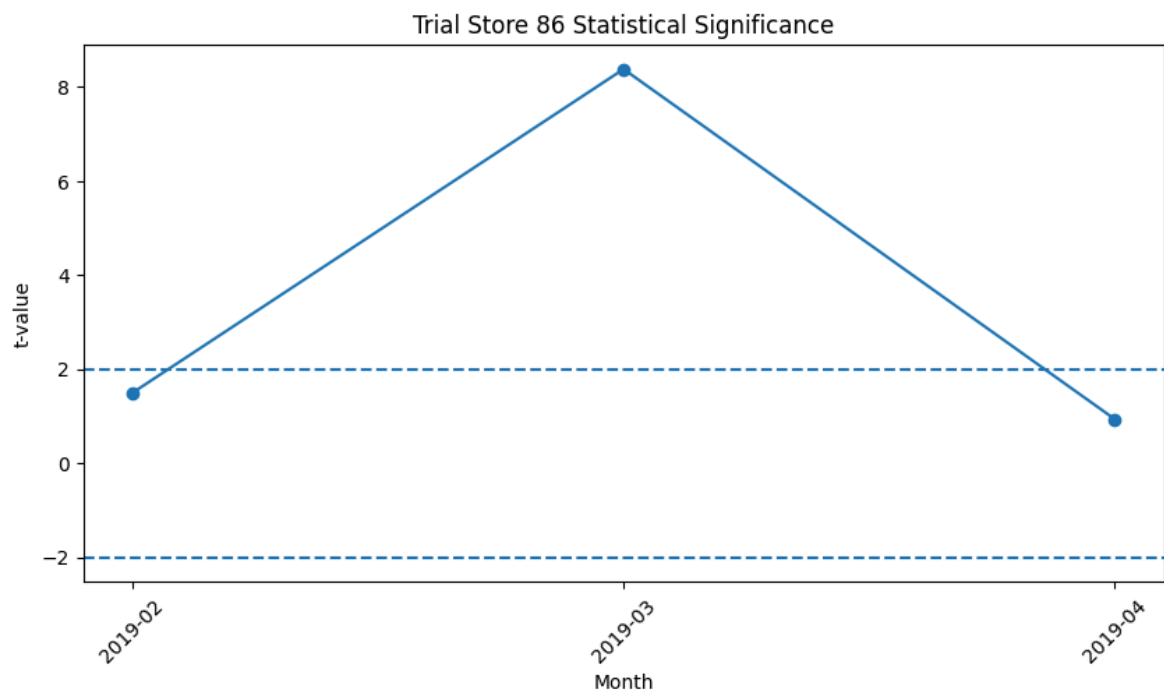
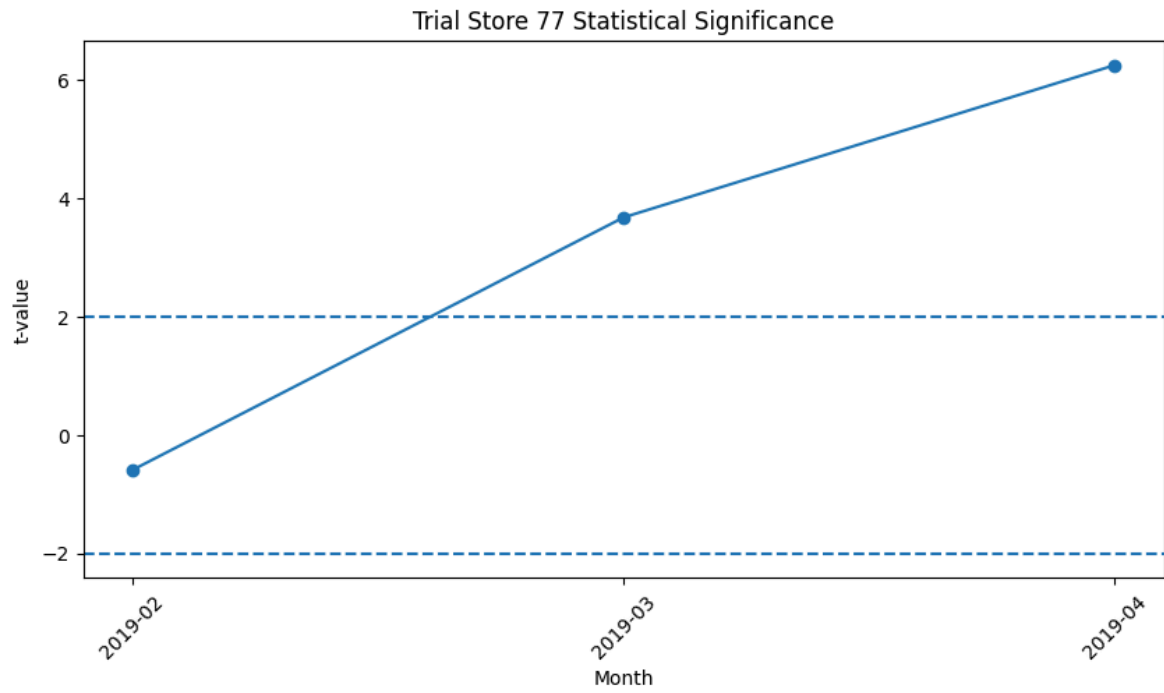
    plt.title(f'Trial Store {trial_store} Statistical Significance')
    plt.xlabel('Month')
    plt.ylabel('t-value')

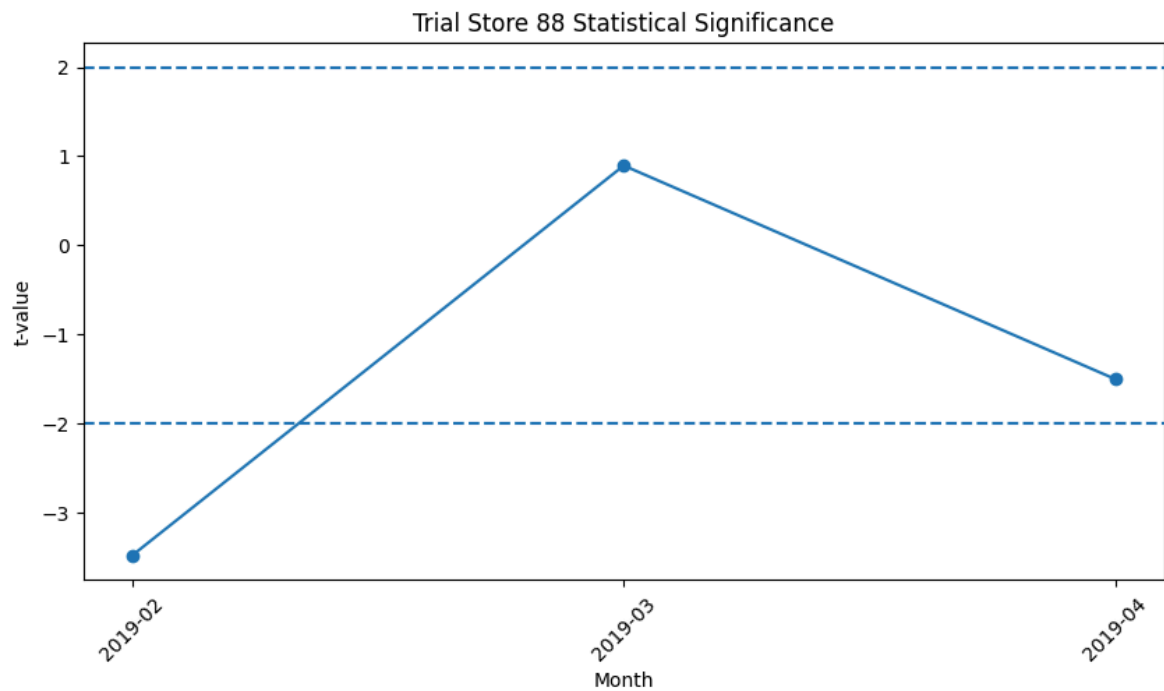
    plt.xticks(rotation=45)
    plt.savefig(
        f'{trial_store}_significance.png',
        dpi=300,
        bbox_inches='tight'
    )

    plt.show()
```

In [50]:

```
plot_trial_significance(sig_77, 77)
plot_trial_significance(sig_86, 86)
plot_trial_significance(sig_88, 88)
```





Statistical Significance Results

A t-value greater than 2 indicates that trial-period uplift exceeded normal historical variation and may therefore be considered statistically significant.

Key findings:

- Store 77 demonstrated statistically significant uplift during March and April 2019.
- Store 86 showed statistically significant uplift only during March 2019, indicating weaker evidence of sustained trial impact.
- Store 88 showed a statistically significant sales decline relative to control in February 2019, suggesting the new layout may have initially disrupted customer behaviour.
- The March 2019 uplift for Store 86 was exceptionally strong ($t=8.4$) and warrants further investigation to confirm it reflects genuine trial impact rather than a data anomaly.

These results suggest the trial layout was most successful in Store 77.

18. Customer Driver Analysis

To understand what contributed to trial performance, customer counts and transactions per customer were reviewed during the trial period.

This helps determine whether sales uplift was driven by more customers visiting the store or by existing customers purchasing more frequently.

```
In [56]: driver_summary = monthly_store_metrics_complete[
    monthly_store_metrics_complete['STORE_NBR'].isin([77, 86, 88, 233, 155, 178]
    ]
    # Pre-trial averages for same stores
    driver_summary_pretrial = driver_summary[
        driver_summary['MONTH_ID'] < '2019-02'
```

```

]

driver_summary_trial = driver_summary[
    (driver_summary['MONTH_ID'] >= '2019-02') &
    (driver_summary['MONTH_ID'] <= '2019-04')
]

driver_summary_trial.groupby('STORE_NBR')[[
    'unique_customers',
    'transactions_per_customer'
]].mean()

```

Out[56]:

	unique_customers	transactions_per_customer
STORE_NBR		
77	47.333333	1.040426
86	109.000000	1.235704
88	128.666667	1.253563
155	96.000000	1.261077
178	112.666667	1.291494
233	38.333333	1.045370

Customer Driver Interpretation

Store 77's uplift appears to be sales-led rather than strongly driven by a large increase in customer count, as its customer base remained relatively small compared with the control group.

Store 86 had higher customer activity during the trial period, which supports the positive March uplift result.

Store 88 had strong customer numbers but did not show positive sales uplift, suggesting that customer volume alone did not translate into stronger sales performance.

19. Trial Outcome Summary

The trial assessment combined correlation analysis, scaled control benchmarking, uplift measurement and statistical significance testing to evaluate the effectiveness of the new store layouts.

Key findings:

- Store 77 demonstrated the strongest and most consistent positive uplift during the trial period.
- Store 86 showed moderate evidence of improvement, although uplift was not consistently significant across all trial months.

- Store 88 did not show statistically significant positive uplift and therefore does not provide strong evidence that the new layout improved sales.

Overall, the trial results suggest that the layout implementation was most successful in Store 77.

```
In [52]: final_results = pd.DataFrame({
    'Trial Store': [77, 86, 88],
    'Control Store': [233, 155, 178],
    'Average Trial Uplift (%)': [
        trial_uplift_77['uplift_percentage'].mean(),
        trial_uplift_86['uplift_percentage'].mean(),
        trial_uplift_88['uplift_percentage'].mean()
    ]
})

final_results
```

Out[52]:

	Trial Store	Control Store	Average Trial Uplift (%)
0	77	233	31.016467
1	86	155	13.580947
2	88	178	-4.785340

	Trial Store	Control Store	Average Trial Uplift (%)
0	77	233	31.016467
1	86	155	13.580947
2	88	178	-4.785340

20. Final Recommendations

Based on the trial analysis:

- The layout trial for Store 77 should be considered successful and may warrant rollout consideration to similar stores.
- Store 86 showed some positive signs of improvement, however additional testing may be beneficial before wider implementation.
- Store 88 showed a net negative average sales effect during the trial and should not be considered for wider rollout of this layout format.

Further investigation into store-specific customer behaviour, demographics and operational differences may help explain variation in trial performance across stores.